Database Fundamentals

Relational Algebra (continuation)

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This presentation contains adapted materials from different sources, including our designated textbook, to be used for educational purposes only

Summary

- Renaming
- Relational Operations: Theta JOIN
- Relational Operations: EQUIJOIN
- NATURAL JOIN
- DIVISION
- Operator decomposition
- DIVISION Decomposition Example

Relational Operators Review:

- Selection **σ** (sigma)
- Projection **π** (pi)
- Union by **U**
- Intersection **∩**
- Set-difference **–**
- Cartesian (cross) product **x**

Schema Example: Example of auction of cars by drivers

DRIVERS

d_id	dname	rating	age

BIDS

d_id	c_id	date	price

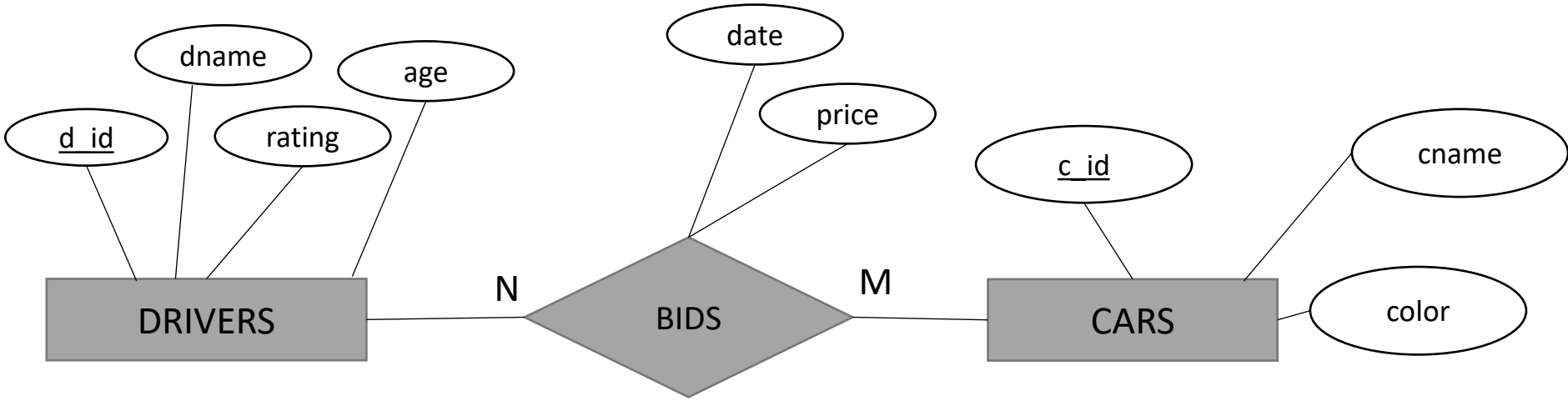
CARS

c_id	cname	color

Drivers (d_id: int, dname: varchar, rating: int, age: numeric);

Cars (c_id:int, cname: varchar, color: varchar);

Bids (d_id: int, c_id: int, day: date, price: numeric);



A CAR can receive bids from many DRIVERS

A DRIVER can bid on many CARS

Rename Operation

- Denoted by **ρ** (Rho)
- Allow us to rename a relation and its members

$$\rho\ C(A1,\ldots,An)\ (E)$$

- Relation E will be renamed to C together with all its fields

Renaming

- Renames D1 relation to BESTDRIVERS (keeps same attributes' names):

$$\rho\ (\text{BESTDRIVERS}, \text{D1})$$

- Renames D1 relation and create BESTDRIVERS containing only dname and rating:

$$\rho\ (\text{BESTDRIVERS}, \pi\ (\text{dname, rating})\ (\text{D1}))$$

CARS

c_id	cname	color

- Renames relation and its attributes:

$$\rho\ (\text{REDCARS}\ (\text{car_id, car_name, car_color})\ (\sigma\ \text{color='Red'}\ (\ \text{CARS}\ )\ )$$

- Renames relation and specific position fields:

$$\rho\ (\ C\ (1 \rightarrow \text{car_id}, 3 \rightarrow \text{car_color})\ (\text{CARS})\ )$$

Renaming

- Necessary for clarity in operations with relations having same field names
- Example: in D x R1, **d_id repeats** so how do we reference one from the other?

RESULT1 = D X R1

d_id	dname	rating	age	d_id	c_id	date
22	Dustin	7	45	22	101	2018-10-10
22	Dustin	7	45	58	103	2018-11-12
95	George	8	55.5	22	101	2018-10-10
95	George	8	55.5	58	103	2018-11-12
58	Rusty	10	35	22	101	2018-10-10
58	Rusty	10	35	58	103	2018-11-12
28	Harry	9	35	22	101	2018-10-10
28	Harry	9	35	58	103	2018-11-12
44	Larry	5	35	22	101	2018-10-10
44	Larry	5	35	58	103	2018-11-12

Renaming

D x R1: (d_id: int, dname: varchar, rating: int, age: numeric,
d_id: int, c_id: int, day: date);

$$\rho\,(C\,(1 \rightarrow d_id1,\ 5 \rightarrow d_id2)\ (D \times R1))$$

- Renames D x R1 to C
- Renames field in position 1 of D x R1 to d_id1 and in position 2 to d_id2

C: (d_id1: int, dname: varchar, rating: int, age: numeric,
d_id2: int, c_id: int, day: date);

Relational Operations: Theta JOIN

- Denoted by ⋈ (bowtie symbol)

$$R \bowtie_{theta} S = D$$

D = R joined with S with condition theta

- **Combines a Cartesian product followed by a selection process**
  - A Join operation pairs two tuples from different relations, if and only if a given join condition is satisfied

- **Very frequently used to implement a relationship between tables**

Theta JOIN Example: Find drivers which driver ID is less that the cars' ID

RESULT (**same schema as cross product**):

d_id	dname	rating	age	d_id	c_id	date
22	Dustin	7	45	58	103	2018-11-12
28	Harry	9	35	58	103	2018-11-12
44	Larry	5	35	58	103	2018-11-12

$$D \bowtie_{D.d_id < R1.d_id} R1$$

Instance of D

d_id	dname	rating	age
22	Dustin	7	45
95	George	8	55.5
58	Rusty	10	35
28	Harry	9	35
44	Larry	5	35

Instance R1 of Bids (R)

d_id	c_id	date
22	101	2018-10-10
58	103	2018-11-12

RESULT1 = D X R1

d_id	dname	rating	age	d_id	c_id	date
22	Dustin	7	45	22	101	2018-10-10
22	Dustin	7	45	58	103	2018-11-12
95	George	8	55.5	22	101	2018-10-10
95	George	8	55.5	58	103	2018-11-12
58	Rusty	10	35	22	101	2018-10-10
58	Rusty	10	35	58	103	2018-11-12
28	Harry	9	35	22	101	2018-10-10
28	Harry	9	35	58	103	2018-11-12
44	Larry	5	35	22	101	2018-10-10
44	Larry	5	35	58	103	2018-11-12

Relational Operations: EQUIJOIN

Type of join were only equal condition operators are used

$$D \bowtie_{D.d_id = R1.d_id} R1 = D \bowtie_{d_id} R1$$

Resulting relation's schema keeps both of the equalty fields

EQUIJOIN Example

$$D \bowtie_{D.d_id = R1.d_id} R1$$

RESULT (**schema keeps both equality fields**):

d_id	dname	rating	age	c_id	d_id	date
22	Dustin	7	45	101	22	2018-10-10
58	Rusty	10	35	103	58	2018-11-12

Expressed as:

$$D \bowtie_{d_id} R1$$

Instance of D

d_id	dname	rating	age
22	Dustin	7	45
95	George	8	55.5
58	Rusty	10	35
28	Harry	9	35
44	Larry	5	35

Instance R1 of Bids (R)

d_id	c_id	date
22	101	2018-10-10
58	103	2018-11-12

RESULT1 = D X R1

d_id	dname	rating	age	d_id	c_id	date
22	Dustin	7	45	22	101	2018-10-10
22	Dustin	7	45	58	103	2018-11-12
95	George	8	55.5	22	101	2018-10-10
95	George	8	55.5	58	103	2018-11-12
58	Rusty	10	35	22	101	2018-10-10
58	Rusty	10	35	58	103	2018-11-12
28	Harry	9	35	22	101	2018-10-10
28	Harry	9	35	58	103	2018-11-12
44	Larry	5	35	22	101	2018-10-10
44	Larry	5	35	58	103	2018-11-12

NATURAL JOIN

- Another variation of JOIN (sub-type of EQUIJOIN)
- Denoted by * or the bowtie symbol by itself 
- Its use **implies that there is at least a field in R with the same name and type of another field in S**
- Connects two relations by equating attributes of the same name, and Projecting out one copy of each pair of equated attributes.

NATURAL JOIN

- In a NATURAL JOIN we do not need to specify the join condition(s). **They are implied**
- Example:

$$RESULT \leftarrow D * R1$$

- Here we imply that there is one or more equal named fields in D and R1 and that we want to join the two tables by those fields
- **Schema keeps only one of the equality fields**

NATURAL JOIN Example

RESULT: D and R1 are joined implicitly by their common attribute d_id

d_id	dname	rating	age	c_id	date
22	Dustin	7	45	101	2018-10-10
28	Rusty	10	35	103	2018-11-12

$D \bowtie R1$

Instance of D

d_id	dname	rating	age
22	Dustin	7	45
95	George	8	55.5
58	Rusty	10	35
28	Harry	9	35
44	Larry	5	35

Instance R1 of Bids (R)

d_id	c_id	date
22	101	2018-10-10
58	103	2018-11-12

RESULT1 = D X R1

d_id	dname	rating	age	d_id	c_id	date
22	Dustin	7	45	22	101	2018-10-10
22	Dustin	7	45	58	103	2018-11-12
95	George	8	55.5	22	101	2018-10-10
95	George	8	55.5	58	103	2018-11-12
58	Rusty	10	35	22	101	2018-10-10
58	Rusty	10	35	58	103	2018-11-12
28	Harry	9	35	22	101	2018-10-10
28	Harry	9	35	58	103	2018-11-12
44	Larry	5	35	22	101	2018-10-10
44	Larry	5	35	58	103	2018-11-12

DIVISION

Denoted by **/** or ÷

R/S:

- **Attributes of S must be a subset of attributes from R**
- The relation returned by the division operator will have **attributes** that are equal to (all attributes of R – all Attributes of S)
- The relation returned by the division operator will return **those tuples from relation R which are associated to every S's tuple(s).**

DIVISION Example R/S = T

R

A	B
a1	b1
a2	b1
a2	b0
a4	b1
a1	b2
a3	b2
a2	b3
a3	b3
a4	b3
a1	b4
a2	b4
a3	b4

÷

S

A
a3

=

T

B
b2
b3
b4

- In this example, I am looking for the **b's in R that are associated with a3**
- Attribute from S is a subset of attributes from R
- The relation returned by the division operator will have attributes that are equal to (all attributes of R – all Attributes of S)
  - b1 is associated with a1, a2, and a4, but not a3
  - **b2 is associated with a1, a2, and a3 (match)**
  - **b3 is associated with a2 and a3 (match)**
  - **b4 is associated with a1, a2, and a3 (match)**

DIVISION Example 2

COURSE_TAKEN

sID	Course
123	COP4710
123	COP3337
128	COP4710
124	COP4710
124	COP2010
125	COP4710
126	COP4710
125	COP3337

÷

COURSES

Course
COP4710
COP3337

=

T

sID
123
125

- sID **123** is associated with **COP4710 & COP3337 (match)**
- sID 128 is associated with COP4710
- sID 124 is associated with COP4710 & COP2010
- sID **125** is associated with **COP4710 & COP3337 (match)**
- sID 126 is associated with COP4710

- In this example, I am looking for **the sID's in COURSES that are associated with COP4710 & COP3337**
- Attributes of COURSE_TAKEN must be a subset of attributes from COURSES
- The relation returned by the division operator will have attributes that are equal to (all attributes of COURSE_TAKEN – all Attributes of COURSES)

DIVISION Example 3

R

A	B
a1	b1
a2	b1
a3	b1
a4	b1
a1	b2
a3	b2
a2	b3
a3	b3
a4	b3
a1	b4
a2	b4
a3	b4

÷

S

A
a1
a2
a3

=

T

B
b1
b4

b1 is associated with **a1, a2, a3,** & a4 (match)
b2 is associated with a1, & a3
b3 is associated with a2, a3, & a4,
b4 is associated with **a1, a2, a3 (match)**

- In this example, I am looking for **the b's in R that are associated with a1 AND a2 AND a3**
- Attributes of R must be a subset of attributes from S
- The relation returned by the division operator will have attributes that are equal to (all attributes of R – all Attributes of S)

Operator decomposition

- Any of the other original relational algebra operations can be expressed as a sequence of other operations. For example:

- **JOIN** can be express as:

$$R \bowtie_{\text{<condition>}} S \equiv \sigma_{\text{<condition>}} (R \times S)$$

- **INTERSECTION** can be express as:

$$R \cap S \equiv (R \cup S) - ((R - S) \cup (S - R))$$

- **DIVISION** can be expressed as :

$$R / S = \pi_x (R) - \pi_x ((\pi_x (R) \times S) - R)$$

DIVISION Decomposition Example

COURSE_TAKEN (R)

sID(X)	Course(Y)
123	COP4710
123	COP3337
128	COP4710
124	COP4710
124	COP2010
125	COP4710
126	COP4710
125	COP3337

÷

COURSES (S)

Course(Y)
COP4710
COP3337

Step 1: $(\pi_X (R) \times S)$

$\pi_X (R)$

sID(X)
123
128
124
125
126

X

COURSES (S)

Course(Y)
COP4710
COP3337

=

$(\pi_X (R) \times S)$

R.Y	S.Y
123	COP4710
123	COP3337
128	COP4710
128	COP3337
124	COP4710
124	COP3337
125	COP4710
125	COP3337
126	COP4710
126	COP3337

$$R / S = \pi_X (R) - \pi_X ((\pi_X (R) \times S) - R)$$

DIVISION Decomposition Example

COURSE_TAKEN (R)

sID(X)	Course(Y)
123	COP4710
123	COP3337
128	COP4710
124	COP4710
124	COP2010
125	COP4710
126	COP4710
125	COP3337

÷

COURSES (S)

Course(Y)
COP4710
COP3337

Step 2: $(\pi_X(R) \times S) - R$

$(\pi_X(R) \times S)$

R.Y	S.Y
123	COP4710
123	COP3337
128	COP4710
128	COP3337
124	COP4710
124	COP3337
125	COP4710
125	COP3337
126	COP4710
126	COP3337

-

COURSE_TAKEN (R)

sID	Course
123	COP4710
123	COP3337
128	COP4710
124	COP4710
124	COP2010
125	COP4710
126	COP4710
125	COP3337

=

$(\pi_X(R) \times S) - R$

R.X	S.Y
128	COP3337
124	COP3337
126	COP3337

$\pi_X((\pi_X(R) \times S) - R)$

R.X
128
124
126

$$R / S = \pi_X(R) - \pi_X((\pi_X(R) \times S) - R)$$

DIVISION Decomposition Example

COURSE_TAKEN (R)

sID(X)	Course(Y)
123	COP4710
123	COP3337
128	COP4710
124	COP4710
124	COP2010
125	COP4710
126	COP4710
125	COP3337

÷

COURSES (S)

Course(Y)
COP4710
COP3337

Step 3: $\pi_X(R) - \pi_X((\pi_X(R) \times S) - R)$

$\pi_X(R)$

R.X
123
128
124
125
126

−

$\pi_X((\pi_X(R) \times S) - R)$

R.X
128
124
126

=

R / S

R.X
123
125

$$R / S = \pi_X(R) - \pi_X((\pi_X(R) \times S) - R)$$